



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO JOTHI AGENCY

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

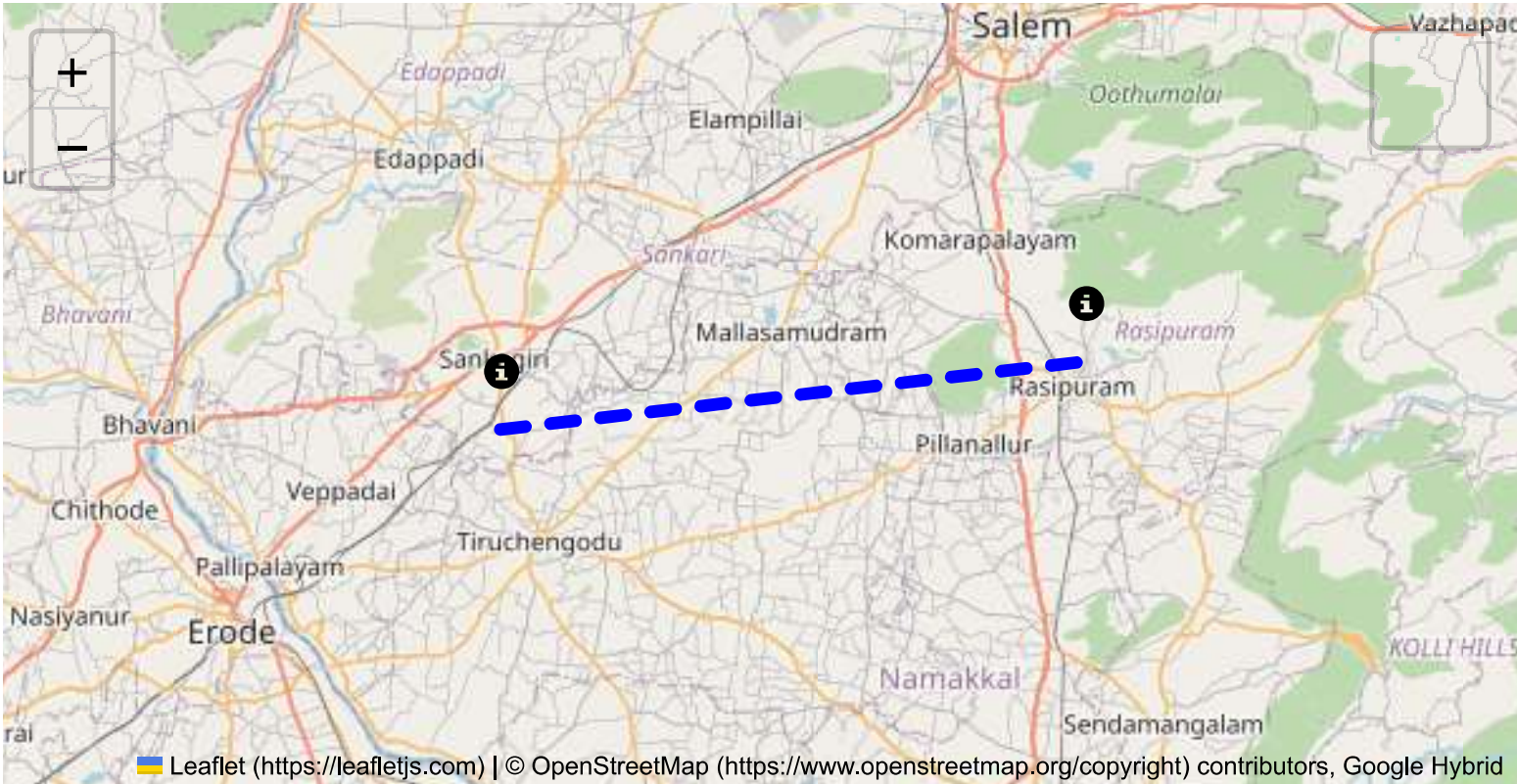
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

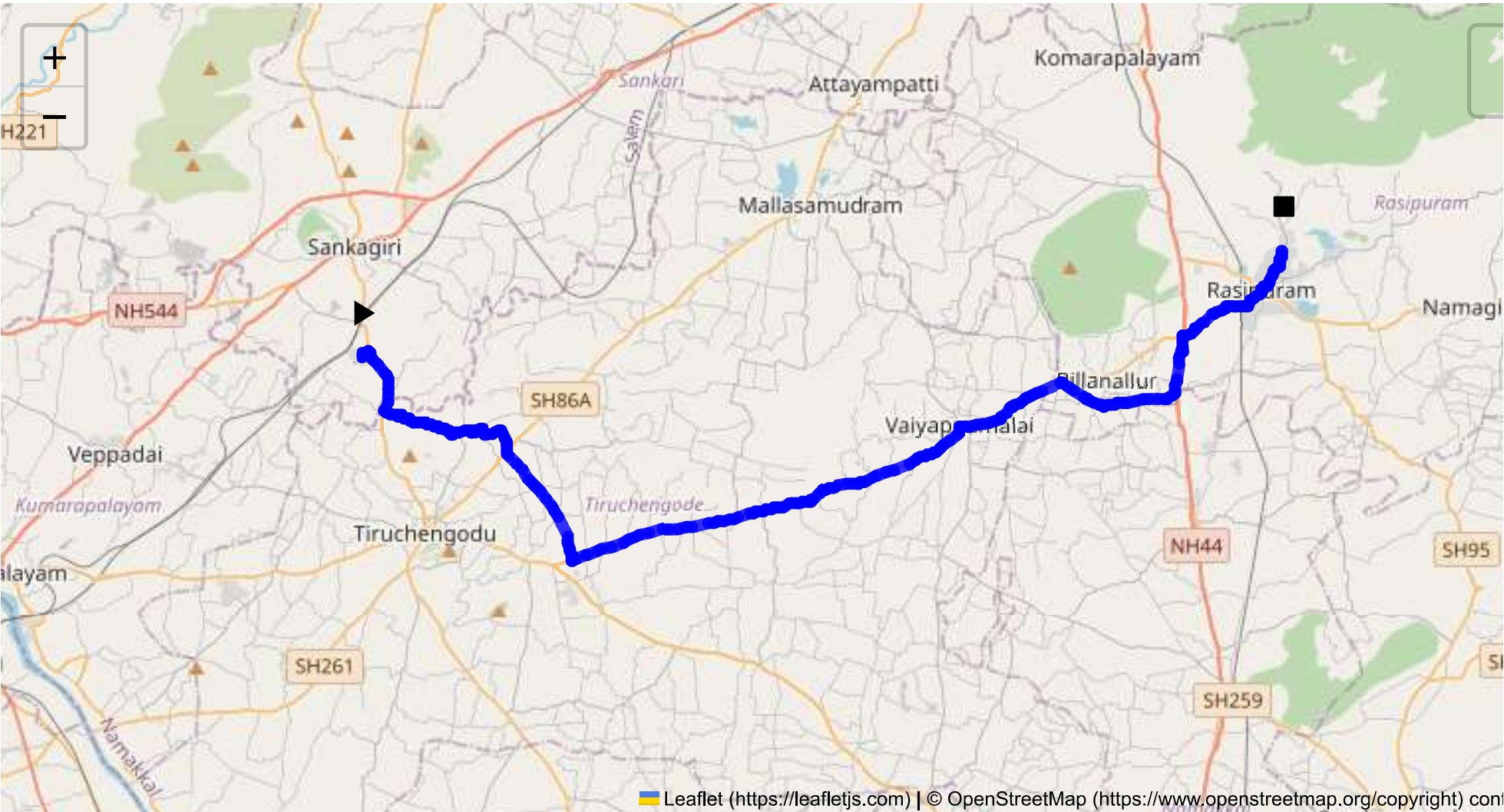
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 46.29 km
Estimated Duration: 1.1 hours
Adjusted Duration (Heavy Vehicle): 1.4 hours
Start: (11.4381, 77.8734)
End: (11.474473, 78.192753)



Welcome to the Journey Risk Management Study

Route Safety Analysis: Sangagiri to Rasipuram, Tamil Nadu

1. Overview of the Route Map

The route spans approximately 46.29 kilometers, mostly following state highways and connecting roads. It runs from Sangagiri, a key industrial area, to Rasipuram, traversing through rural and semi-urban regions.

The roads are a mix of moderate traffic density, with varying road widths.

2. Typical Weather Conditions and Potential Weather-Related Hazards

- **Weather:** The region experiences a tropical climate. Winters are mild, while summers can be hot, with temperatures soaring in May and June.
- **Rainfall:** Significant rainfall occurs during the monsoon (June to September) and the northeast monsoon (October to December).
- **Hazards:**
 - Monsoon rains can cause localized flooding and waterlogging.
 - Reduced visibility due to heavy rains can affect driving conditions.

3. Analysis of Traffic Patterns

- **Peak Hours:** Morning (8:00 AM - 10:00 AM) and evening (5:00 PM - 7:00 PM) witness higher traffic.
- **Congestion-Prone Areas:** Town centers and major junctions, particularly in and around Sangagiri and during entry into Rasipuram.

4. Assessment of Road Quality and Infrastructure

- Roads are generally well-maintained but can have potholes, especially post-monsoon.
- Some stretches have narrow shoulders, increasing difficulty for large vehicles.
- Bridges and culverts need to be checked for height and weight restrictions.

5. Suggestions for Alternative Routes for Emergencies

- **Alternative Route:** In case of a blockade, consider NH-44 as an alternative for bypassing congested or hazardous sections.
- Familiarize with local roads connecting villages that might serve as detours.

6. Summary of Local Regulations Affecting Hazardous Material Transport

- Permits required for transport of hazardous materials.
- Movement of such vehicles might be restricted during peak urban hours.
- Specific restrictions on passing through populated areas to minimize risk.

7. Overview of Historical Incidents

- No major historical incidents reported involving heavy vehicles or hazardous materials on this specific route.
- General caution advised due to scattered minor accidents associated with overspeed and unawareness of local turnings.

8. Environmental Considerations and Sensitive Areas

- The route passes through agricultural zones.
- Environmental and noise pollution should be minimized to avoid disturbing villages and farms.
- No major ecological reserves or protected areas along this specific stretch.

9. Analysis of Communication Coverage

- Generally good cell coverage throughout the area.
- Potential dead zones in mountainous or dense vegetative areas should be noted; ensure drivers are aware and prepared.

10. Estimated Emergency Response Times

- Urban areas: ~30-45 minutes due to accessibility.
- Rural stretches: ~45-60 minutes; longer if immediate medical or fire services are required.
- Encourage vehicle operators to have emergency contacts pre-saved.

12. Overall Summary of Risk Assessment

- **Weather Risks:** Moderate to high during monsoon.
- **Traffic Risks:** Moderate, due to congestion in urban areas and unexpected stops in rural zones.
- **Road Risks:** Moderate; need for awareness while navigating narrow or less maintained sections.
- **Regulatory Risks:** High, specifically related to permits and restricted movement times for hazardous materials.

Recommendations

- Conduct regular route checks post-monsoon for road conditions.
- Pre-plan journeys to avoid peak times and ensure vehicle fitness, especially brakes and tires.
- Maintain a robust communication protocol to preempt coverage-related issues.
- Align with local authorities for updated regulations and emergency strategies.

By considering these conditions, the safety of transporting hazardous materials on this route can be significantly increased.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.46141, 78.18540	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
1	Turn	Medium	11.43822, 77.87348	30 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
4	Turn	High	11.41981, 77.88078	15 KM/Hr
5	Turn	Medium	11.41876, 77.88487	30 KM/Hr
6	Turn	Medium	11.41829, 77.88568	30 KM/Hr
7	Turn	High	11.41859, 77.88692	15 KM/Hr
8	Turn	Medium	11.41761, 77.88739	30 KM/Hr
9	Turn	Medium	11.41716, 77.88935	30 KM/Hr
10	Turn	Medium	11.41630, 77.89077	30 KM/Hr
11	Turn	Medium	11.41601, 77.89198	30 KM/Hr
12	Turn	Medium	11.41603, 77.89479	30 KM/Hr
13	Turn	Medium	11.41439, 77.90025	30 KM/Hr
14	Turn	High	11.41377, 77.90073	15 KM/Hr
15	Turn	Medium	11.41375, 77.90128	30 KM/Hr
16	Turn	Medium	11.41389, 77.90161	30 KM/Hr
17	Turn	Medium	11.41325, 77.90313	30 KM/Hr
18	Turn	Medium	11.41276, 77.90349	30 KM/Hr
19	Turn	High	11.41230, 77.90485	15 KM/Hr
20	Turn	Medium	11.41262, 77.90506	30 KM/Hr
21	Turn	Medium	11.41273, 77.90524	30 KM/Hr
22	Turn	High	11.41329, 77.90978	15 KM/Hr
23	Turn	High	11.41424, 77.91433	15 KM/Hr
24	Turn	Medium	11.41409, 77.91458	30 KM/Hr
25	Turn	High	11.41208, 77.91551	15 KM/Hr
26	Turn	Medium	11.41218, 77.91651	30 KM/Hr
27	Turn	High	11.41347, 77.92096	15 KM/Hr
28	Turn	Medium	11.41336, 77.92130	30 KM/Hr
29	Turn	Medium	11.40833, 77.92362	30 KM/Hr
30	Turn	High	11.40566, 77.92355	15 KM/Hr
31	Turn	High	11.40557, 77.92343	15 KM/Hr
32	Blind Spot	Blind Spot	11.36920, 77.94568	10 KM/Hr
33	Turn	Medium	11.41343, 78.08073	30 KM/Hr
34	Turn	High	11.41466, 78.08070	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
35	Turn	Medium	11.41473, 78.08082	30 KM/Hr
36	Turn	High	11.42962, 78.11578	15 KM/Hr
37	Turn	Medium	11.42450, 78.15278	30 KM/Hr
38	Turn	High	11.42622, 78.15562	15 KM/Hr
39	Turn	High	11.44547, 78.15837	15 KM/Hr
40	Turn	Medium	11.45587, 78.18057	30 KM/Hr
41	Turn	Medium	11.45597, 78.18073	30 KM/Hr
42	Turn	High	11.46035, 78.18534	15 KM/Hr
43	Turn	Medium	11.46215, 78.18568	30 KM/Hr
44	Turn	Medium	11.46262, 78.18724	30 KM/Hr
45	Turn	Medium	11.46273, 78.18738	30 KM/Hr
46	Turn	High	11.46331, 78.18767	15 KM/Hr
47	Turn	Medium	11.46342, 78.18797	30 KM/Hr
48	Turn	Medium	11.46989, 78.19186	30 KM/Hr
49	Turn	Medium	11.47170, 78.19180	30 KM/Hr
50	Turn	Medium	11.47357, 78.19262	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
8	hospital	Sudha Hospital	11.4555773, 78.1760125	30 km/h	Medium
9	hospital	Government Hospital Blood Bank, Rasipuram	11.4612082, 78.1849249	30 km/h	Medium
10	hospital	Government Hospital, Rasipuram	11.4612987, 78.1851369	30 km/h	Medium
11	hospital	M.R. Hospital, Namakkal	11.4613462, 78.1861702	30 km/h	Medium
12	hospital	dr.thangavel dch hospital	11.4619955, 78.1849181	30 km/h	Medium
13	police	Police Station Rasipuram	11.4617825, 78.1847411	30 km/h	Medium
14	hospital	surya hospital	11.4622163, 78.1840544	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
15	hospital	Rasi Hospital	11.4618368, 78.1855695	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	marketplace	Farmer Market	11.4146849, 78.0805463	30 km/h	Medium
1	school	Government Higher Secondary School	11.4150045, 78.0838172	30 km/h	Medium
2	school	Government school	11.4248027, 78.1323902	30 km/h	Medium
3	college	Paavai Arts And Science College For Women	11.4262443, 78.1546987	30 km/h	Medium
4	school	SRV EXCEL Matriculation Higher Secondary School	11.4299679, 78.152616	30 km/h	Medium
5	college	Thiruvalluvar Govt arts college	11.4449352, 78.1595562	30 km/h	Medium
6	school	Venkateswara School	11.4468258, 78.1595761	30 km/h	Medium
7	school	Venkateswara Middle school	11.4472063, 78.1596191	30 km/h	Medium
16	school	Government Hr Sec School	11.4654519, 78.1869866	30 km/h	Medium
17	school	Government Hr Sec School	11.4654357, 78.186985	30 km/h	Medium
18	school	RC Secard Heart Hr. Sec. School	11.4697866, 78.1911116	30 km/h	Medium
19	school	RC Secard Heart Hr. Sec. School	11.4698476, 78.1910744	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46141, 78.18540



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41981, 77.88078



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41876, 77.88487



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41829, 77.88568



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41859, 77.88692



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41761, 77.88739



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41716, 77.88935



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41630, 77.89077



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41601, 77.89198



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41603, 77.89479



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41439, 77.90025



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41377, 77.90073



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41375, 77.90128



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41389, 77.90161



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41325, 77.90313



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41276, 77.90349



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41230, 77.90485



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41262, 77.90506



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41273, 77.90524



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41329, 77.90978



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41424, 77.91433



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41409, 77.91458



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41208, 77.91551



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41218, 77.91651



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.41347, 77.92096



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41336, 77.92130



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.40833, 77.92362



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.40566, 77.92355



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.40557, 77.92343



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.36920, 77.94568



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41343, 78.08073



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41466, 78.08070



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41473, 78.08082



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.42962, 78.11578



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.42450, 78.15278



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.42622, 78.15562



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44547, 78.15837



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45587, 78.18057



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45597, 78.18073



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.46035, 78.18534



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46215, 78.18568



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46262, 78.18724



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46273, 78.18738



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46331, 78.18767



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46342, 78.18797



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46989, 78.19186



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.47170, 78.19180



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.47357, 78.19262